📘 AZ-104 – LAB 04

TASK 3 & TASK 4 – DETAILED NOTES (SECURITY + DNS)

🔐 TASK 3: CONFIGURE COMMUNICATION BETWEEN ASG AND NSG

🎯 Objective of Task 3

The goal of Task 3 is to **secure network traffic inside the Virtual Network** by:

- Controlling **which resources can communicate**
- Restricting **which ports are allowed**
- Blocking **unnecessary internet access**
- Making security rules **scalable and readable**

In real Azure environments, **network security is enforced at the subnet and NIC level**, not inside applications.

🧠 Why Azure Uses NSG and ASG

Azure separates **security enforcement** and **workload grouping**:

Component	Purpose
NSG	Enforces traffic rules
ASG	Groups workloads logically

This separation makes security **scalable and maintainable**.

🔹 Network Security Group (NSG) – Deep Explanation

An **NSG** is a **stateful firewall** that filters traffic using rules.

Key properties of NSG rules:

- Evaluated **by priority**
- Lower number = **higher priority**
- First matching rule is applied

- Evaluation stops after a match

Default NSG behavior:

- **Inbound:** Deny all
- **Outbound:** Allow all

This means:

If you don't explicitly allow traffic, it is blocked.

🔹 Application Security Group (ASG) – Deep Explanation

An **ASG** groups **VMs by application role**, not IP address.

Why ASG is important:

- IP addresses can change
- VM scale sets add/remove instances
- ASGs stay consistent even when infrastructure changes

Example:

asg-web → Web servers

asg-db → Database servers

Security rules reference **roles**, not IPs.

🧱 What We Implemented in Task 3

1️⃣ Created ASG: asg-web

Purpose:

- Represents web-tier workloads
- Used as **source** in NSG rules

At this stage:

- No VM is attached
- Still valid and expected

2️⃣ Created NSG: myNSGSecure

Purpose:

- Acts as subnet-level firewall
- Controls inbound and outbound traffic

3️⃣ Associated NSG with SharedServicesSubnet

This is a **critical step**.

Why?

- NSG only works **after association**
- Subnet-level NSG affects **all resources inside it**

🔐 Inbound Rule – Allow Web Traffic from ASG

Rule logic:

Allow traffic FROM asg-web

ON ports 80 and 443

USING TCP

Why ports 80 and 443?

- 80 → HTTP
- 443 → HTTPS
- Most web applications use these ports

Why Priority = 100?

- Ensures rule is evaluated **before default deny**
- Leaves space for future rules
- Follows industry best practice

🚫 Outbound Rule – Deny Internet Access

Rule logic:

Deny ALL outbound traffic

TO Internet

ON all ports

Why deny outbound internet?

- Prevents malware communication
- Prevents data exfiltration
- Forces traffic through controlled paths (firewall, proxy)

Why destination ports = 0–65535?

- Means **all ports**
- Equivalent to *

Why Priority = 4096?

- Higher than inbound allow rules
- Lower than default allow (65001)
- Overrides default outbound internet access

🔍 Final Security Result (Task 3)

✔ Only allowed traffic enters subnet
✔ Internet access is blocked
✔ Security is role-based
✔ Rules are scalable and readable

🌐 TASK 4: CONFIGURE PUBLIC AND PRIVATE AZURE DNS

🎯 Objective of Task 4

The goal of Task 4 is to enable **name resolution**, so resources can be accessed using:

hostname.domain.com

instead of:

IP addresses

DNS is **mandatory** for real-world cloud environments.

🔹 Why DNS is Critical in Azure

Without DNS:

- Applications break when IP changes
- Configurations become fragile
- Scaling becomes difficult

With DNS:

- Applications use names
- Infrastructure changes are hidden
- Environments scale smoothly

🌍 Public DNS Zone – Deep Explanation

What Public DNS does:

- Resolves names **from the internet**
- Used for:
  - Websites
  - APIs
  - Public services

Example:

www.contosolab.com → Public IP

🧱 Public DNS Record (A Record)

An **A record** maps:

hostname → IPv4 address

Example:

www.contosolab.com → 10.1.1.4

(Used as demo value in lab)

🔐 Private DNS Zone – Deep Explanation

Private DNS:

- Works **only inside VNets**
- Is not accessible from the internet
- Used for internal services

Example:

sensorvm.private.contosolab.com → 10.1.1.4

🔗 Virtual Network Link – Why It's Required

Private DNS zones do NOTHING until linked.

The link:

- Tells Azure **which VNet can use this DNS**
- Enables name resolution inside that network

Without linking:
❌ DNS records exist
❌ But cannot be resolved

📃 Private DNS Recordsets – Why We Added Them

A **recordset** defines:

"When someone asks for this name, return this IP."

Without recordsets:

- DNS zone exists
- But resolves nothing

Recordsets make DNS **functional**.

🔍 Final DNS Result (Task 4)

✔ Public names resolve externally
✔ Private names resolve internally
✔ VNets use DNS without exposing services
✔ IP changes won't break applications

🎯 INTERVIEW-READY MASTER ANSWERS

Q: Why use ASG instead of IP addresses?
A: ASGs group workloads logically and remain stable as infrastructure scales.

Q: Why is NSG stateful?
A: Return traffic is automatically allowed once a connection is established.

Q: Why deny outbound internet traffic?
A: To reduce attack surface and prevent data exfiltration.

Q: Why use private DNS instead of public DNS internally?
A: To keep internal services hidden from the internet.

Q: Why link private DNS to VNet?
A: Without linking, name resolution does not work.

✅ FINAL KEY TAKEAWAY (IMPORTANT)

Task 3 secures how traffic flows
Task 4 controls how resources are found

Together, they form the **core of Azure network architecture**.